

Xin Zhao, Ph.D.
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Education

2015 – 2021	University of Alabama <i>Ph.D. in Computer Science</i> Major Field of Study: Software Engineering Advisor: Dr. Jeff Gray
2014 – 2015	Wuhan University <i>Graduate Study</i> Major Field of Study: Software Engineering Advisor: Dr. Ruimin Hu
2009 – 2013	Hebei Normal University <i>Bachelor of Science in Computer Science</i> Major Field of Study: Software Engineering

Academic Appointments

Sep. 2021 – Present	Seattle University Assistant Professor, Department of Computer Science
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Grants and Fellowships

Summer 2026,	Faculty Fellow
Summer 2025,	Summer Undergraduate Research Program
Summer 2024,	College of Science and Engineering, Seattle University
Summer 2023	Summer 2026, Project: <i>Narrative Role Bias in Generative AI: How Large Language Models Assign Agency, Harm, Blame, and Survival Across Social Identities</i> . Amount: \$7,000. Role: PI. Summer 2025, Project: <i>Is Artificial Intelligence an Elixir to the Software Engineering Community? An Empirical Study Among Product Managers and Leaders</i> . Amount: \$6,500. Role: PI. Summer 2024, Project: <i>Self-Portrait of Software Developers Where Software is Not at The Forefront</i> . Amount: \$6,500. Role: PI. Summer 2023, Project: <i>The Climate Surrounding Workplace Discrimination in Human Resources: An Empirical Study in STEM</i> . Amount: \$6,000. Role: PI.

Jan 2026 – June 2026

Faculty Fellow

Technology Ethics Initiative Student-Scholar Program

College of Science and Engineering, Seattle University

Project: *Artificial Intelligence Tools for K-12 School Leaders: A Professional Development Perspective Among Administrators in Washington State*. Amount: \$2,500. Role: PI.

2022 – 2023

Faculty Fellow

The Center for Business Ethics

Albers School of Business and Economics, Seattle University

Project: *An Empirical Study of Employment Discrimination in Information Technology Corporates*. Amount: \$7,000. Role: PI.

Grants Under Review

NSF STEM K-12

Project: *Accelerating Computational Capability and Expanding Subsidized Systems (ACCESS): A Systematic Improvement Pathway for K-8 Computing Education*, Amount: \$750,000, Role: PI.

Publications

Journals

Ana Carolina de Souza Mendes, Mason Adsero, Joshua Palicka, Nurulla Zholdoshev, and **Xin Zhao**. "AI-crafted narratives: an empirical study on generating interactive stories using generative pre-training transformers." *Applied Intelligence*, vol. 55, no. 15 (2025): 1004. <https://doi.org/10.1007/s10489-025-06833-3>

Saheed Popoola, **Xin Zhao**, Jeff Gray, and Antonio Garcia-Dominguez, "Classifying changes to LabVIEW and Simulink models via changeset metrics." *Innovations in Systems and Software Engineering* (2024): 1-15. <https://doi.org/10.1007/s11334-024-00577-Y>.

Xin Zhao, Gurshan Rai, and Saheed Popoola, "Ask or tell: An empirical study on modeling challenges from LabVIEW community", *Journal of Computer Languages*, 80 (2024): 101284. <https://doi.org/10.1016/j.cola.2024.101284>.

Xin Zhao and Jeff Gray, "Towards a Metrics Suite for the Complexity Analysis of LabVIEW Systems Models," *Science of Computer Programming*, vol. 227, article no. 102931. (2023): 1 – 26. <https://doi.org/10.1016/j.scico.2023.102931>.

Saheed Popoola, **Xin Zhao**, and Jeff Gray, "Evolution of Bad Smells in LabVIEW Graphical Models," *Journal of Object Technology*, vol. 20, no. 1 (2021): 1-15. <http://dx.doi.org/10.5381/jot.2021.20.1.a1>

Book Chapter

Xin Zhao, Jiang Zhe, and Jeff Gray, "Text Classification and Topic Modeling for Online Discussion Forums - An Empirical Study from the Systems Modeling Community," *Trends and Applications of Text Summarization Techniques*. Idea Group Inc., ISBN: 978-1522593737, 2020, Chapter 6, pp. 151 – 186. <https://doi.org/10.4018/978-1-5225-9373-7.ch006>

Hei Chio, Li Huang, and **Xin Zhao**, “Peer-to-Peer Fraud in China: Geographical Variations,” accepted for presentations at the 2026 *American Society of Criminology Meeting*, Nov 2026, Chicago, USA.

Xin Zhao, Brian Vu, and Sitesh Pattanaik, “Is Artificial Intelligence an Elixir to the Software Engineering Community? An Empirical Study Among Managers,” in *Proceedings of the 3rd ACM International Conference on AI-Powered Software (AIware)*, pp. 168-177, *ACM SIGSOFT Distinguished Paper Award*, July 2026, Montreal, Canada. <https://doi.org/10.1145/3805760.3814907>.

Xin Zhao and Parth Manish Reshamwala, “Personalized Medicine Meets Artificial Intelligence: A Systematic Literature Review,” in *Proceedings of the 3rd International Conference on AI-Health*, pp. 82-89. March 2026, Valencia, Spain.

Xin Zhao and Ai Sun, “Mapping K–12 educators’ professional learning pathways to AI literacy and integration: An empirical study,” in *Proceedings of the 18th International Conference on Computer Supported Education (CSEDU)*, vol. 2, pp. 1278–1289, 2026, Benidorm, Spain. <https://doi.org/10.5220/0014980400004021>.

Xin Zhao and Narissa Tsuboi, “Early-Career Software Developers – Are you Sinking or Swimming?” in *Proceedings of the 46th International Conference on Software Engineering (ICSE)*, pp. 166-176, April 2024, Lisbon, Portugal. <https://doi.org/10.1145/3639475.3640106>.

Julian Stefanick and **Xin Zhao**. "Popular Songs: The Sentiment Surrounding the Conversation." In *Proceedings of the 19th International Conference on Advanced Data Mining and Applications (ADMA)*, pp. 368-382, August 2023, Shenyang, China. https://doi.org/10.1007/978-3-031-46661-8_25

Xin Zhao and Riley Young, “Workplace Discrimination in Software Engineering: Where We Stand Today,” in *Proceedings of the 45th International Conference on Software Engineering (ICSE)*, pp. 188-193, May 2023, Melbourne, Australia. <https://doi.org/10.1109/icse-seis58686.2023.00026>

Saheed Popoola, **Xin Zhao**, Jeff Gray, Antonio Garcia-Dominguez, “Classifying Changes to Models via Changeset Metrics,” in *Proceedings of 14th System Analysis and Modelling Conference (SAM)*, pp. 276–285, October 2022, Montreal, Canada. <https://doi.org/10.1145/3550356.3561563>

Xin Zhao, Jeff Gray, and Taylor Riché, “A Survey-Based Empirical Evaluation of Bad Smells in LabVIEW Systems Models,” in *Proceedings of 28th Conference on Software Analysis, Evolution, and Reengineering (SANER)*, pp. 177-188, March 2021, Hawaii, USA. <https://doi.org/10.1109/SANER50967.2021.00025>

Xin Zhao and Jeff Gray, “BESMER: An Approach for Bad Smells Summarization in Systems Models,” in *Proceedings of the 22nd International Conference on Model Driven Engineering Languages and Systems (MoDELS)*, pp. 304-313, September 2019, Munich, Germany. <https://doi.org/10.1109/MODELS-C.2019.00047>

Xin Zhao and Jeff Gray, “Design Guidelines for Feature Model Construction: Exploring the Relationship between Feature Model Structure and Structural Complexity,” in *Proceedings of 7th International Conference on Model-Driven Engineering and Software Development, (MODELSWARD)*, pp. 323 – 331, February 2019, Prague, Czech Republic. <https://doi.org/10.5220/0007388703230331>

Xin Zhao, Austin Payne, and Travis Atkison “TTEXTS: A Dynamic Framework to Reverse UML Sequence Diagrams from Execution Traces,” in *Proceedings of 16th International Conference on Software Engineering Research and Practice, (SERP)*, pp. 82 – 88, July 2018, Las Vegas, Nevada, USA,

Xin Zhao, “Feature-oriented Modeling for Collaborative Virtual Environment Construction,” *Doctoral Symposium at ACM/IEEE 20th International Conference on Model Driven Engineering Languages and Systems (MoDELS)*, pp. 494 – 499, September 2017, Austin, Texas, USA.

Invited Talks

Guest Speaker, “Empirical Software Engineering: From Discipline to Interdiscipline,” *CS Seminar Series, Department of Computer Science, Seattle University*, October 2022.

Guest Lecture, “Bad Smells in Model-Based Systems Engineering: An Empirical View,” *CS 573 Advanced Software Engineering, Department of Computer Science, Boise State University*, October 2021.

Guest Speaker, “A Complexity Analysis of LabVIEW Systems Models,” *2020 ACM Mid-Southeast Conference* (online conference), November 2020.

Experience

Sep. 2021 – Present

Assistant Professor, Seattle University

Conduct research in Empirical Software Engineering, Artificial Intelligence, and K-12 Computing Education; teach undergraduate and graduate software engineering courses.

Aug. 2015 – May 2021

Doctoral Research, University of Alabama

Summarization, empirical evaluation, and identification of code smells in systems models.

<i>Jul. 2020 – Jul. 2021</i>	Research Assistant, University of Alabama Department of Education – EIR : Assisted in year-long “CS Professional Development” experiences offered to teachers who receive instruction from evidence-based curricula that form a CS course pathway.
<i>Jan. 2020 – Jun. 2020</i>	Research Assistant, University of Alabama NSF STEM+C (#1639971): Designed lesson plans for the Edison robots for K-12 schools in the Alabama Black Belt region.
<i>Aug. 2015 – Dec. 2019</i>	Teaching Assistant, University of Alabama CS 104 – Computer Science Principles, Fall 2015; Spring 2016, 2017. CS 201 – Data Structures and Algorithms, Spring 2018. CS 300 – Operating Systems, Fall 2018, 2019; Spring 2019. CS 495 – Capstone Computing, Fall 2016, 2017; Spring 2017, 2018.
<i>Jun. 2018 – Aug. 2018</i>	Developer, Gorgas Library at the University of Alabama Developed and integrated components of FOLIO, an open-source library services platform, for Gorgas Library at the University of Alabama.
<i>Sep. 2014 – Jul. 2015</i>	Graduate Student Researcher, National Engineering Research Center for Multimedia Software at Wuhan University Developed and optimized surveillance systems for person identification with machine learning techniques.
<i>Sep. 2011 – Aug. 2012</i>	Co-founder and Manager, BBS for School of Software, Hebei Normal University Co-founded and managed the official bulletin board system for the School of Software at Hebei Normal University, overseeing its maintenance for one year.
<i>Feb. 2012 – Aug. 2012</i>	Undergraduate Student Researcher, Key Laboratory of Internet of Things of Hebei Province, China Designed and developed the <i>Tanlers</i> Energy Management System.
K-12 Outreach	Lead Instructor , Equity in Computer Science Education Program (ECSEP), Seattle University, 2025, 2026. Coordinator , Game On event at 2021 Alabama State Science Olympiad (SO), March 2021. Judge , University of Alabama STEM showcase, March 2021. Team Member , Exploring Computer Science (ECS) Online Summer Professional Development, July 2020. Team Member , UTeach AP Computer Science Principles (AP CSP) Online Professional Development, July 2020.

Administrator and Head Judge, Annual Alabama Robotics Contest, April 2019.

Assistant, CS4HS Summer Camp, June 2019 and June 2018.

Assistant, Annual Alabama Robotics Contest, April 2018, 2017, 2016.

Course Assistant, Computer Science for High School (CS4HS, sponsored by Google), July 2016.

Awards and Honors

Awards

ACM Travel Grants to MoDELS Conference, Amount: \$500, 2017.

Runner-up, 2015 International Mathematical Contest in Modeling (MCM)/ Interdisciplinary Contest in Modeling (ICM), 2015.

Academic Scholarship

Graduate Student Fellowship, Wuhan University, 2014-2015.

Academic Excellence Scholarship, Hebei Normal University, 2012.

Excellent Student Award, Hebei Normal University, 2011.

National Encouragement Scholarship, Hebei Normal University, 2010.

University Service

Committee

Master of Science in Artificial Intelligence Program Calendar Unification Committee, Computer Science Department, Seattle University, Academic Year 2025 - 2026

Tenure-Track Faculty Search Committee, Computer Science Department, Seattle University, Academic Year 2022-2024

Graduate Admission Committee, Computer Science Department, Seattle University, Academic Year 2023 – present

Club Organization

SU AWS Cloud Club, founder and advisor, 2025 - present

SU Badminton Club, founder and advisor, 2025 - present

Students (graduation date)

Master

Ai Sun (current)

Timothy Pranoto (June 2026)

Rishi Munuswamy (June 2025)

Yash Atre (June 2025)

Parth Reshamwala (June 2025)

Jasmine Pletcher (December 2024)

Sai Supraj Reddy Madhavapeddi (June 2024)

Narissa Tsuboi (June 2024)

Rupansh Phutela (June 2023)

Sonali Desarda (Mar 2023)

<i>Undergraduate</i>	Shaan Gada (current) Brian Vu (current) Riley Young (June 2024) Gurshan Rai (June 2024) Julian Stefanick (December 2023)
<i>K-12</i>	Raehan Baliga (current, Tesla STEM High School)

Courses

<i>Seattle University</i>	CPSC 2430, Data Structures CPSC 4260/5260, Refactoring and Software Design CPSC 5110, Fundamentals of Software Engineering CPSC 5200, Software Architecture and Design CPSC 5890, Seminar
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Professional Service

<i>Program Committee</i>	Annual Modeling and Simulation Conference (ANNSIM 2026, 2025, 2024)
<i>Journal Reviewer</i>	Journal of Software and Systems Modeling Transactions on Software Engineering and Methodology Software: Practice and Experience
<i>Conference Reviewer</i>	International Conference on Model Driven Engineering Languages and Systems (MoDELS), 2020. European Conference on Modelling Foundations and Applications (ECMFA), 2020, 2019. International Conference on Model Transformation (ICMT), 2017. International Conference of Evaluation and Modelling Methods for Systems Analysis and Development (EMMSAD), 2017.
<i>Student Volunteer</i>	International Conference on Model Driven Engineering Languages and Systems (MoDELS), 2019, 2017.